Taylor & Francis
Taylor & Francis Group

Transactions of the American Fisheries Society

ISSN: 0002-8487 (Print) 1548-8659 (Online) Journal homepage: www.tandfonline.com/journals/utaf20

A Genetic Assessment of Ohio River Walleyes

Matthew M. White, Todd W. Kassler, David P. Philipp & Scott A. Schell

To cite this article: Matthew M. White, Todd W. Kassler, David P. Philipp & Scott A. Schell (2005) A Genetic Assessment of Ohio River Walleyes, Transactions of the American Fisheries Society, 134:3, 661-675, DOI: 10.1577/T03-218.1

To link to this article: https://doi.org/10.1577/T03-218.1

Published online: 09 Jan 2011.

Submit your article to this journal

Article views: 156

View related articles

Citing articles: 5 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=utaf20

Transactions of the American Fisheries Society 134:661–675, 2005
© Copyright by the American Fisheries Society 2005
DOI: 10.1577/T03-218.1

[Article]

A Genetic Assessment of Ohio River Walleyes

MATTHEW M. WHITE*

Department of Biological Sciences, Ohio University, Athens, Ohio 45701, USA

TODD W. KASSLER[1] AND DAVID P. PHILIPP

Center for Aquatic Ecology, Illinois Natural History Survey, 607 East Peabody Drive, Champaign, Illinois 61820, USA

SCOTT A. SCHELL

Ohio Division of Wildlife, 360 East State Street, Athens, Ohio 45701, USA

Abstract.—Walleyes *Sander vitreus* from nine Ohio River pools were examined by means of two independent restriction fragment length polymorphism (RFLP) analyses of mitochondrial DNA, allozyme analysis of nuclear DNA, and microsatellite analysis of nuclear DNA. The primary objective of this project was to document genetic variation and population structuring in walleyes among the nine pools analyzed. Both RFLP analyses revealed the presence of a distinct haplotype that was most abundant in walleyes collected from two tailwaters in the upper Ohio River (river miles 54 and 84), suggesting a remnant native population. The allozyme analysis also revealed differentiation between walleyes with the two haplotypes, even though there were no fixed differences. The microsatellite analysis indicated highly significant structuring among the nine pools analyzed. A lack of microsatellite differentiation between the two haplotypes detected in the same pool suggests that the fish are freely interbreeding. If the unique haplotype is from a remnant native stock of walleyes in the Ohio River, it could serve as a genetic marker to establish a broodstock for stocking walleyes in the upper Ohio River drainage. The second objective of the project was to evaluate the accuracy of field identifications of walleyes, saugers *S. canadensis*, and their hybrid, saugeyes, and to determine the direction of hybridization. Twenty-seven percent of the fish identified as walleyes were misidentified (10% were F_X hybrids and 17% F_1 hybrids). All F_1 hybrids were from walleye females and sauger males.

Fisheries biologists recognize that stocking practices can have negative impacts on native stocks (Ryman and Utter 1987). The introduction of nonnative individuals or different species has resulted in the decline or loss of fitness of native stocks through intraspecific introgression and interspecific hybridization (Campton 1987; Philipp et al. 2002).

The Ohio River Fisheries Management Team is a multistate agency group that has identified the management of walleye *Sander vitreus*, sauger *S. canadensis*, and saugeye (female walleye × male sauger) to be a priority in the Ohio River because of their recreational value. Walleyes and saugers are both native to the Ohio River, which has been extensively stocked with Lake Erie walleyes. Saugeyes also occur naturally in the Ohio River in low numbers but are locally abundant owing to consistent stocking in several Ohio River tributaries, including the Muskingum River, which receives annual stockings. In addition, most on-stream impoundments that were traditionally stocked with Lake Erie strain walleyes are now being stocked with saugeyes produced from female walleyes of Lake Erie origin and male Ohio River saugers.

Although an allozyme study by White and Schell (1995) identified significant differentiation among walleye populations in the Ohio River, no geographic structure was detected. The study also detected limited backcrossing between introduced saugeyes and walleyes or saugers. Billington (1996) detected several divergent haplotypes in the Ohio River and suggested that they were relics of prestocking walleye populations. Stepien and Faber (1998) also analyzed walleyes in the Ohio River using DNA sequence analysis and identified several haplotypes. One group of haplotypes was consistent with walleyes from Lake Erie. A second haplotype was highly divergent from the Lake Erie haplotype, suggesting a separation of up to 1.5 million years (Stepien and Faber 1998). The pres-

* Corresponding author: whitem@ohio.edu

[1] Present address: Washington Department of Fish and Wildlife, Genetics Laboratory, 600 Capitol Way North, Olympia, Washington 98501-1091, USA.

Received December 10, 2003; accepted September 23, 2004
Published online May 5, 2005

TABLE 1.—Sample sizes of walleyes from nine tailwaters on the Ohio River and Lake Erie, Pymatuning Lake (Pennsylvania), Salt Fork Lake (Ohio), and Lake Winnebago (Wisconsin) for the allozyme, two RFLP, and microsatellite DNA analyses.

Location	River mile[a]	Analysis			
		Allozyme	RFLP1	RFLP2	Microsatellite
New Cumberland	54	30	29	36	36
Pike Island	84	20	21	26	26
Willow Island	162	39	14	21	21
Racine	238	4	3	8	8
Byrd	280			5	5
Greenup	342	40	14	40	39
Melhdahl	436	9	6	19	19
Markland	606	16	14	20	18
McAlpine	720	6	4	7	7
Lake Erie		30	5		
Lake Winnebago		30			
Pymatuning Lake				12	12
Salt Fork Lake				4	4

[a] 1 mi = 1.61 km.

ence of the Lake Erie haplotypes in the Ohio River is probably the result of past stocking of Lake Erie walleyes into the Ohio River (White and Schell 1995).

The primary objective of this study was to identify stock structure by collecting walleye samples from the Ohio River and subjecting them to analyses using three molecular techniques. This work expands that of White and Schell (1995) to include more riverwide sampling (river miles [RM] 54.4–720, measuring from the confluence at Pittsburg; 1 mi = 1.61 km) using two independent restriction fragment length polymorphism (RFLP) analyses of mitochondrial DNA (mtDNA), an allozyme analysis of nuclear DNA, and a microsatellite analysis of nuclear DNA. This expanded study is part of an Ohio River Fisheries Management Team project undertaken to evaluate stock structure in Ohio River walleye populations and to develop shared management strategies. The second objective of this study was to use allozyme analysis to confirm field identifications of walleyes followed by an RFLP analysis to determine the nature and direction of the hybridization events.

Methods

Samples.—Walleyes, saugers, and saugeyes were obtained from nine tailwaters along the Ohio River (Table 1) during 1998 and 1999. Samples were also collected from Lake Winnebago, Wisconsin, Lake Erie, Salt Fork Lake, Ohio, and Pymatuning Lake, Pennsylvania, as reference samples. Salt Fork Lake is a walleye broodstock lake for the state of Ohio. The walleyes in this lake are derived from Lake Erie. Pymatuning Lake is the source of walleyes used for stocking throughout western Pennsylvania. These walleyes are also Lake Erie derived. Fish were collected during lockchamber surveys or by electrofishing. No attempt was made to select fish from a particular size- or age-class. Samples were placed on wet ice in the field and then stored at −20°C until analysis was conducted. All fish analyzed by means of allozymes and in the first RFLP analysis (RFLP1) (except those from Lake Erie and Lake Winnebago) were used in the second RFLP analysis (RFLP2) and the microsatellite DNA survey. When available, additional specimens were used in RFLP2 and the microsatellite DNA survey.

Allozymes.—Tissue samples of white skeletal muscle, eye, and liver were dissected, homogenized in 100 mM tris–HCl (pH 7.0), and centrifuged at 22,000 × gravity for 10 min at 4°C. Tissue extracts were subjected to vertical starch gel electrophoresis coupled with histochemical staining as described by Philipp et al. (1979) with modifications from Koppelman and Philipp (1986). A total of 15 loci were surveyed to assess the genetic variation in walleyes from nine pools along the Ohio River (Table 2). Three loci that exhibit fixed differences between walleyes and saugers (*sAAT-2**, *mMDH-B**, and *PGM**) were used to identify the sampled fish as walleyes, saugers, or F_1 or F_X hybrids. Fish homozygous for walleye or sauger alleles at all three diagnostic loci were considered pure walleyes or saugers, respectively. The F_1 hybrids possessed genotypes that were heterozygous at all three diagnostic loci. The F_X hybrids possessed some combination of homozygous and heterozygous genotypes. When one is using genotypes at only three diagnostic loci, it is possible for some fish to be misidentified. Only fish identified as walleyes based on allozyme genotype were used in the subsequent RFLP and microsatellite analyses. *F*-statistics and genic heterogeneity were estimated with BIOSYS-1 (Swofford and Selander 1981). Rogers' (1972) genetic distance coefficients were used to determine phenetic relationships among populations using a cluster analysis involving the unweighted pair-group method with arithmetic averages (Sneath and Sokal 1973).

Mitochondrial DNA.—Total genomic DNA was extracted from white muscle using either a modified hexadecyltrimethylammonium bromide method (Fields et al. 1989) or the high-salt modification technique (Hillis et al. 1996), depending

TABLE 2.—Eleven enzyme systems (15 loci) analyzed for walleyes, saugers, and saugeyes. Enzyme numbers are those recommended by IUBMB (1992), and locus designations follow Shaklee et al. (1990). Buffer systems are described in Koppelman and Philipp (1986); TC = tris-citrate.

Enzyme number	Enzyme name	Locus	Tissue analyzed	Buffer system
2.6.1.1	Aspartate amino transferase	*sAAT-2**	Liver	TC
		*mAAT**	Liver	TC
1.1.1.1	Alcohol dehydrogenase	*ADH**	Liver	TC
2.7.4.3	Adenlyate kinase	*AK-1**	Muscle	TC
	Calcium binding protein	*CBP**	Muscle	TC
3.1.3.11	Fructose 1,6-bisphosphatase	*FBP**	Muscle	TC
1.1.1.42	Isocitrate dehydrogenase	*sIDHP-1**	Muscle	TC
		*sIDHP-2**	Liver	TC
1.1.1.27	Lactate dehydrogenase	*LDH-C**	Eye	TC
1.1.1.37	Malate dehydrogenase	*sMDH-A**	Muscle	TC
		*sMDH-B**	Muscle	TC
		*mMDH**	Muscle	TC
1.1.1.44	Phosphogluconate dehydrogenase	*PGDH**	Liver	TC
5.4.2.2	Phospho-glucomutase	*PGM**	Muscle	TC
1.15.1.1	Superoxide dismutase	*SOD**	Liver	TC

on the laboratory. The DNA was resuspended in distilled water.

Two different RFLP assays were used. The first RFLP analysis was performed on a segment of the mitochondrial genome near the control region or 12S RNA region using the L15998-PRO and CSBDH primers (Alvarado Bremer et al. 1998). Amplification was performed in an MJ Research PTC-100-60 thermocycler under the following conditions: 2 min at 95°C, followed by 29 cycles of 45 s at 94°C, 1 min at 45°C, and 2.5 min at 70°C. Samples of each amplification product were digested with each of six restriction endonucleases (*Bst*U I, *Dde* I, *Dpn* I, *Hae* I, *Hha* I, and *Msp* I). Digestion fragments were separated electrophoretically in a gel consisting of 2% agarose and 1% synergel (Diversified Biotech) in tris–acetate–EDTA buffer with 0.5μg of ethidium bromide/mL.

The second RFLP analysis used the left domain of the mitochondrial control region (from the proline tRNA gene to the central conserved section) using the universal primers L15926 (Kocher et al. 1989) and H16498 (Meyer et al. 1990). Amplification was performed in an MJ Research PTC-100-25 thermocycler under the following conditions: 2 min at 95°C, followed by 30 cycles of 30 s at 94°C, 30 s at 50°C, and 1 min at 72°C. The amplifications produced an approximately 800-base-pair (bp) fragment. Agarose gel electrophoresis (0.8% in tris–boric acid [borate]–EDTA) was used to verify the presence of polymerase chain reaction (PCR) amplification products and to determine whether restriction endonuclease digestion occurred. Walleyes with the Ohio River haplotype contain a diagnostic *Ase* I restriction site (Table 3) that yields two fragments of approximately 550 and 250 bp. Walleyes derived from Lake Erie carry a diagnostic *Bsu*36 I restriction site that yields two fragments of approximately 600 and 200 bp. This analysis also allows the detection of sauger haplotypes since this amplified fragment in sauger does not cut with either restriction enzyme. Thus, no saugers or hybrids with sauger maternal background were included. All samples surveyed by RFLP1 were surveyed by RFLP2 (Table 1).

TABLE 3.—Variable nucleotide positions used to differentiate Lake Erie (G) and Ohio River (O) haplotypes. Positions 125 and 126 correspond to an *Ase* I restriction site (ATTAAT); positions 131 and 132 correspond to a *Bsu*36 I restriction site (CCTNAGG). Sequences are from Stepien and Faber (1998).

	Nucleotide position			
Haplotype	125	126	131	132
Lake Erie	C	C	G	G
Ohio River	A	T	A	A

Microsatellite DNA.—Microsatellite loci were amplified following the methods given in Borer et al. (1999) and Wirth et al. (1999). Five loci were surveyed (*Svi4*, *Svi6*, *Svi17*, *Svi18*, and *SviL5*), and the amplification products were verified on agarose gels. Microsatellite allele sizes were determined using 8% polyacrylamide gels with multiple lanes of a 25-bp ladder as a size standard on each gel. Gels were photographed with a Kodak DC290 digital camera with a dark yellow filter. Gel images were analyzed with the 1D Advanced Gel-Scanning

TABLE 4.—Sample sizes (*N*) and allele frequencies scored for six polymorphic allozyme loci in walleyes from eight tailwaters of the Ohio River (indicated by river mile [RM]), Lake Erie, and Lake Winnebago. The collections from RM 54 and 84 are divided into individuals scored with the Lake Erie (G) haplotype and the Ohio River (O) haplotype.

Locus and allele	Population											
	54G (N = 19)	54O (N = 11)	84G (N = 12)	84O (N = 8)	162 (N = 39)	238 (N = 4)	342 (N = 40)	436 (N = 9)	606 (N = 16)	720 (N = 6)	Lake Erie (N = 30)	Lake Winnebago (N = 30)
*sAAT-2**												
**100*	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.944	1.000	1.000	1.000	1.000
**115*	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.056	0.000	0.000	0.000	0.000
*ADH**												
**60*	0.026	0.000	0.125	0.000	0.077	0.000	0.083	0.056	0.062	0.167	0.259	0.167
**100*	0.974	1.000	0.875	1.000	0.923	1.000	0.917	0.944	0.938	0.833	0.741	0.833
*CBP**												
**100*	0.526	0.227	0.500	0.188	0.551	0.625	0.575	0.500	0.719	0.833	0.552	0.833
**160*	0.474	0.773	0.500	0.813	0.449	0.375	0.425	0.500	0.281	0.167	0.448	0.167
*FBP**												
**85*	0.000	0.000	0.000	0.000	0.000	0.125	0.000	0.000	0.062	0.000	0.000	0.000
**100*	1.000	1.000	1.000	1.000	1.000	0.875	1.000	1.000	0.938	1.000	1.000	1.000
*sIDHP-2**												
**75*	0.500	0.773	0.667	0.688	0.590	0.500	0.444	0.278	0.375	0.333	0.310	0.483
**100*	0.500	0.227	0.333	0.313	0.410	0.500	0.556	0.722	0.625	0.667	0.690	0.517
*sMDH-B**												
**100*	0.500	0.364	0.667	0.250	0.577	0.125	0.662	0.778	0.750	0.417	0.741	0.883
**120*	0.500	0.636	0.333	0.750	0.423	0.875	0.338	0.222	0.250	0.583	0.259	0.117

Program (American Applied Biotechnology, Fullerton, California).

Each individual walleye was categorized as belonging to either the Ohio River stock or the Lake Erie stock based on the sequence-specific RFLP analysis and was genotyped at each microsatellite locus. Individuals were pooled into populations by locality and mtDNA haplotype. The Microsatellite Tool-Kit (Park 2001) was used to estimate allele frequencies and heterozygosities; GENEPOP (Raymond and Rousset 1995) was then used to conduct Fisher's exact test for Hardy–Weinberg equilibrium and population differentiation. Tests of differentiation for all pairs of populations were conducted using fixation indices (Weir and Cockerham 1984) with the program FSTAT (Goudet 1995). An analysis of molecular variance was conducted with Arlequin (Schneider et al. 2000). The values of F_{ST} were plotted against river mile distance to determine whether there was a relationship between genetic differentiation and geographic distance.

Results

Allozyme Analysis

A total of 224 walleyes from eight Ohio River populations, Lake Erie, and Lake Winnebago were surveyed. Eleven enzyme systems (15 loci) were screened, and six of the loci were found to be polymorphic (Table 4). Four of the loci (*ADH**, *CPB**, *sIDHP-2**, and *sMDH-B**) were polymorphic in all populations. Genotype frequencies for walleyes with the two different haplotypes (G and O) at the New Cumberland (RM 54) and Pike Island (RM 84) tailwaters were calculated separately and analyzed as separate groups (i.e., the RM54 G, RM54 O, RM84 G, and RM84 O haplotypes).

For the Ohio River samples, 2 of 31 comparisons with Hardy–Weinberg expectations were significant (*sIDHP-2** at RM54 G and *FBP** at RM606). Significant differentiation ($P < 0.01$) was detected at four loci (*ADH**, *CPB**, *sIDHP-2**, and *sMDH-B**). The F_{ST} values ranged from 0.05 (*sAAT-2**) to 0.148 (*sMDH-B**) and averaged 0.121. The cluster analysis (Figure 1) revealed that walleyes from the RM54 O and RM84 O groups (the samples from New Cumberland and Pike Island with the O haplotype) clustered together and were the most distant from all the other walleye populations. The walleyes from the RM54 G and RM84 G groups (the samples from New Cumberland and Pike Island with the G haplotype) clustered with the other walleye samples from the Ohio River, all of which had the G haplotype. To evaluate the differentia-

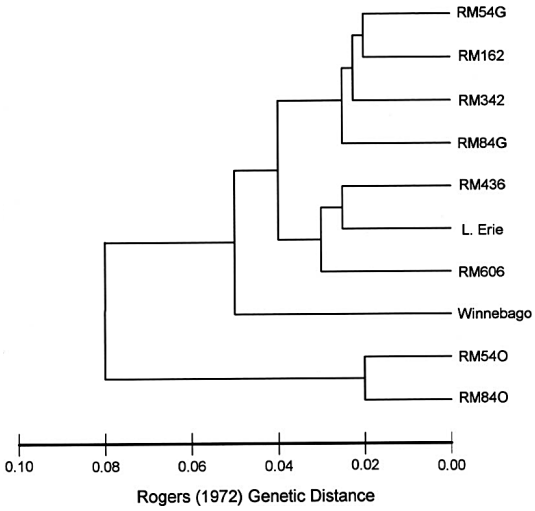

FIGURE 1.—Cluster of genetic distances for populations of walleyes from the Ohio River and Lake Erie produced by the unweighted pair-group method with arithmetic averages. See text for haplotype descriptions.

TABLE 5.—Number of walleyes from nine tailwaters of the Ohio River, Lakes Erie, Pymatuning, and Salt Fork scored with the Lake Erie (G) haplotype and the Ohio River (O) haplotype in RFLP analyses 1 and 2. See Table 1 for more information in sampling locations.

Location	RFLP1		RFLP2	
	G	O	G	O
Lake Erie	5	0		
Pymatuning Lake			12	0
Salt Fork Lake			4	0
New Cumberland	18	11	22	14
Pike Island	12	9	8	18
Willow Island	14	0	17	4
Racine	3	0	8	0
Byrd			5	0
Greenup	14	0	39	1
Meldahl	6	0	19	0
Markland	14	0	19	1
McAlpine	4	0	7	0

tion observed between the co-occurring haplotype populations (RM54 O–RM84 O and RM54 G–RM84 G), we pooled the O and G individuals at each locality and reevaluated departures from Hardy–Weinberg equilibrium. Only one significant departure was observed (*sIDHP-2** at RM54).

RFLP Analyses

Analysis of mtDNA 1.—A total of 112 walleyes from eight Ohio River tailwaters and Lake Erie were assayed. Two of the six restriction endonucleases (*Bst*U I and *Msp* I) that were analyzed revealed banding patterns that were diagnostic between walleyes and saugers, ensuring that no saugers were included in the analysis. The remaining four restriction endonucleases (*Dde* I, *Dpn* II, *Hae* III, and *Hha* I) revealed two composite banding patterns (G and O) that were consistently detected. The G haplotype (the Lake Erie haplotype) was detected in all eight localities along the Ohio River and in Lake Erie. The O haplotype (the Ohio River haplotype) was only detected within the New Cumberland and Pike Island tailwaters (Table 5). The 33 F_X hybrids identified by the allozyme analysis were assayed with the four polymorphic restriction endonucleases and revealed 31 individuals with the G haplotype and two individuals with the O haplotype.

Analysis of mtDNA 2.—Of the 211 walleyes surveyed with this analysis, 38 possessed the O haplotype and 163 possessed the G haplotype (Table 5). The O haplotype was rare or absent at localities in the lower Ohio River. This haplotype increased in frequency in populations in the upper Ohio River. At the New Cumberland and Pike Island tailwaters, the frequency of the O haplotype was 0.45 and 0.72, respectively. The O haplotype was not observed in Salt Fork or Pymatuning lakes.

Microsatellite DNA

The allele sizes detected by the microsatellite analysis were consistent with other published results for walleyes (Borer et al. 1999; Wirth et al. 1999). Considerable variation was observed at all five microsatellite loci (Appendix 1). The numbers of alleles at each locus ranged from 8 (*Svi18*) to 15 (*SviL5*). After adjusting for multiple comparisons (Rice 1989), 20 of 70 comparisons with Hardy–Weinberg expectations were significantly different. All significant departures indicated a deficit of heterozygotes.

Highly significant differentiation among populations of walleyes was observed at all loci ($P < 0.001$). Over 93% of the variance was due to within-population variation. Approximately 7% was due to differences among populations. Less than 1% of the variance was due to differences between haplotypes. No significant differentiation was observed between co-occurring Ohio River and Lake Erie haplotype samples (New Cumberland, Pike Island, and Willow Island). No significant relationship was observed between geographic distance and F_{ST} ($P = 0.51$; Figure 2).

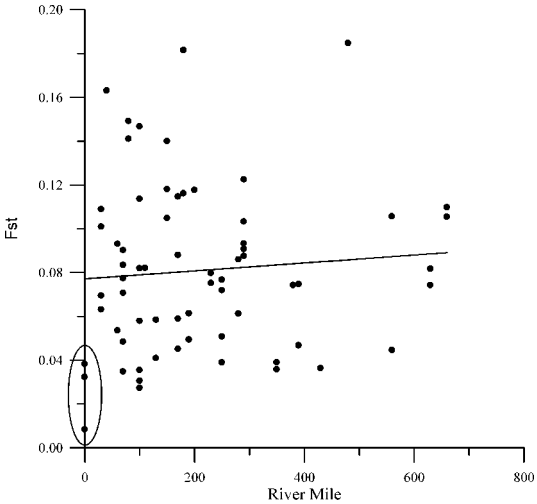

FIGURE 2.—Plot of F_{ST} values and river mile distances for all pairwise comparisons of walleye populations along the Ohio River (slope = 0.0006; P = 0.51). The three comparisons between co-occurring haplotype populations are circled.

Accuracy of Visual Identification of Walleyes

Of the total of 287 individuals visually identified in the field as walleyes, only 211 (73.5%) were confirmed as walleyes by the allozyme analysis. Of the remaining (misidentified) fish, 16 (5.6%) were saugers, 27 (9.4%) were F_1 hybrids, and 33 (11.5%) were F_X hybrids. The largest number of hybrids was observed at Greenup tailwater (RM 342; Kassler and Philipp 2001); F_X hybrids were detected in all areas except New Cumberland and McAlpine (RM 720).

Discussion

Although genetic studies have been conducted on walleyes throughout much of their natural range (e.g., Billington and Hebert 1988; Billington et al. 1992; Billington and Strange 1995; Stepien and Faber 1998; Gatt et al. 2000, 2002), with the exception of a single allozyme study (White and Schell 1995) walleyes from the Ohio River have not been extensively evaluated. Walleyes were originally abundant in the Ohio River but then experienced a serious decline followed by moderate recovery (Trautman 1981). Inland stocking programs by Ohio, Pennsylvania, and West Virginia have introduced large numbers of walleyes of Lake Erie origin into the streams of the Ohio River (J. Navarro, Ohio Department of Natural Resources; C. Obara, West Virginia Department of Natural Resources; and W. Shuey, Pennsylvania Fish and Boating Commission, personal communication). The first objective of this study was to determine whether there is any remnant stock structure to walleye populations in the Ohio River. Several states that border the Ohio River have also stocked fertile saugeyes into Ohio River tributaries since 1980 (White and Schell 1995; Kassler and Philipp 2000). Because saugeyes can successfully backcross with walleyes, their presence has the potential to affect the genetic integrity of native walleyes in the Ohio River.

Assessment of RFLP Differentiation

Two independent analyses of mtDNA identified a haplotype (haplotype O) similar to that detected by Stepien and Faber (1998). Both analyses identified the same walleyes with this haplotype using the restriction endonucleases *Dde* I and *Bsu*36 I. Although this haplotype was observed in five of the nine tailwaters sampled, the highest frequencies occurred in the two most upstream study sites in the Ohio River (the New Cumberland and Pike Island tailwaters; Table 5).

Several hypotheses may be invoked to explain the distribution of the Ohio River (O) haplotype. First, walleyes with this haplotype may be remnants of a native population from the Ohio River that was historically more widespread than currently. A riverwide decline of walleye populations may have reduced their distribution so that now they are only found in the upper reaches of the Ohio River. Alternatively, the O haplotype may be more abundant in the upper Ohio River because it evolved in the Teays River. The upper Ohio River and its Teays River ancestor have a long history of isolation from Great Lakes drainages (Flint 1971). Stepien and Faber (1998) used a sequence analysis to suggest that the two different types of walleye they observed from the Ohio River and Lake Erie were isolated approximately 1.5 million years ago, a time frame that is consistent with a Teays River origin (Thornbury 1965). Recently, Palmer (1999) identified a New River, Virginia, stock of walleyes that may be native to that area of the river. This stock possesses unique mtDNA haplotypes. We recently tested our RFLP2 assay on a small sample of these fish (courtesy of E. Hallerman, Virginia Tech). The New River haplotype was cut with *Ase* I but not with *Bsu*36 I, suggesting that these New River haplotypes and the Ohio River haplotype are related. This would support a Teays River origin for the haplotypes.

The presence of walleyes with both the Ohio River (O) haplotype and the Lake Erie (G) hap-

lotype in the tailwaters of the Ohio River could be a result of upstream or downstream migration by stocked walleyes from Lake Erie–derived stocks. The walleyes that were used for inland stocking in Ohio, Pennsylvania, and West Virginia were historically derived from such stocks. All of the walleye samples analyzed from Lake Erie, Salt Fork Lake, and Pymatuning Lake had the same G haplotype. It is also possible that a haplotype related to the G haplotype was found in fish native to the lower parts of the Ohio River and the adjacent Mississippi River. We did not test that hypothesis. A comparison of walleyes from the lower Mississippi River with those from the lower reaches of the Ohio River would allow the resolution of this question. Natal homing or philopatry may also explain why the Ohio River haplotype exists in high proportion in a small area of the upper part of the river and not the lower part (Todd and Hass 1993; Jennings et al. 1996).

Assessment of Allozyme Differentiation

The allozyme data suggest significant genetic differentiation among populations. White and Schell (1995) also observed significant structuring, but no geographic pattern was detected. The allozyme data also provide evidence that walleyes with the Ohio River and Lake Erie haplotypes are genetically divergent (Figure 1). If there was random mating among individuals with the two different haplotypes, we would predict homogeneity of allele frequencies in areas where the two haplotypes co-occur (New Cumberland and Pike Island). The observed allele frequencies of walleyes with the two mtDNA haplotypes suggest that they are to a large degree mating assortatively. If so, a Wahlund effect should be apparent when the haplotype populations at each locality are pooled. However, when the haplotype samples from New Cumberland and Pike Island were pooled, no additional departures from Hardy–Weinberg equilibrium were observed. This suggests that the two haplotype populations are not significantly different and are not isolated to any significant extent. White and Schell (1995) were not aware of the two haplotypes and thus their analysis was based on pooled samples. A reanalysis of their data did not result in a clustering of New Cumberland and Pike Island samples.

Assessment of Microsatellite Differentiation

Considerable variation was detected within and among populations (Appendix 1). The large number of alleles detected at each of the different loci may be a reflection of the presence of two distinct walleye types in the Ohio River: native walleyes and those introduced from Lake Erie. The large number of alleles also suggests a large effective population size.

The large number of significant departures from Hardy–Weinberg equilibrium expectations (all were heterozygote deficits) may have several causes. Several of the sample sizes were small in comparison with other microsatellite studies (e.g., Nielsen et al. 1999; King et al. 2001) and may not be reflective of the populations in the tailwaters sampled. Further, some of the samples apparently consisted of individuals from the same year-class and thus might be progeny from a relatively few matings. Errors in allele identification may also have contributed to the heterozygote deficits.

A deficit of heterozygotes may also be due to the presence of null alleles. Null alleles arise by mutation in the flanking regions, causing failure of the priming and the absence of that PCR product. Heterozygotes for the null allele would appear as homozygotes for the amplified allele, thus biasing allele and genotype frequency estimates. Because the distribution of heterozygote deficits was relatively equal across loci, the null allele hypothesis is unlikely.

The microsatellite survey did not detect significant differentiation between co-occurring fish with the Lake Erie and Ohio River haplotypes at New Cumberland, Pike Island, and Willow Island. The F_{ST} values for these comparisons were among the lowest observed (Figure 2). This is not consistent with the allozyme results and suggests that there is random mating between individuals with the different haplotypes. The absence of historical (prestocking) data precludes any estimate of the degree of recent introgression between the co-occurring populations. Larger sample sizes at these three localities might help to resolve this lack of distinctiveness between the haplotype populations.

Allozymes are an important group of genetic markers for the analysis of differentiation among fish populations. It is often assumed that allozyme differentiation reflects historical patterns of gene flow and genetic drift (Allendorf and Seeb 2000). Our allozyme data would thus suggest either that there is some degree of assortative mating or that the secondary contact between the haplotype populations is recent enough that there has not been sufficient time to achieve equilibrium. The microsatellite data suggest that the co-occurring haplotype populations are freely interbreeding. When differences between allozymes and microsatellites

are detected, it could be due to differences in their responses to natural selection (Allendorf and Seeb 2000) or rates of mutation, or the increased sensitivity of microsatellites to changes in population structure, size, and dispersal (Estoup et al. 1998). Microsatellites may be more sensitive and permit the detection of differentiation when allozymes do not (Arnaud et al. 2001; Estoup et al. 1998). Larger sample sizes and an examination of additional microsatellite loci would help to resolve this issue and may identify instances of linkage disequilibrium. This would support a hypothesis of secondary contact between the Ohio River and Lake Erie populations. It is also possible that the allozyme results reflect a remnant signal of historical differences whereas the microsatellites reflect recent changes to local demographics, population size, and breeding structure.

Management Implications

Genetic Structure of Walleye Populations

This genetic evaluation confirms that a divergent walleye haplotype is present in the Ohio River and that it is most abundant in the New Cumberland and Pike Island tailwaters. These two tailwaters are separated from the direct effects of stocking programs that occur both upstream and downstream by two dams. Because the upper Ohio River is affected by cool, mountain runoff and is characterized by slightly higher gradient than the middle and lower sections, one might expect to find a remnant population of native walleyes in this reach.

The Ohio River has been altered in relatively recent times by two sets of navigational dams, which not only limited fish movement but also eliminated riffle habitat and inundated traditional spawning sites. Navigational and commercial dredging operations have greatly reduced shallow gravel deposits and available spawning sites as well. Despite these gross habitat alterations, the Ohio River walleye haplotype persists, demonstrating its adaptability.

Introgression of Walleyes and Saugeyes

Natural walleye × sauger hybrids are rare in the Ohio River. These fertile hybrids coexist with sympatric populations of both parental species. Isolation mechanisms in their spawning are believed to be responsible for limiting genetic introgression, since both parental species are broadcast spawners. Walleyes are known to spawn before saugers, but both species have been documented to use the same spawning sites (G. L. Siegwarth, Iowa Department of Natural Resources, unpublished).

The natal homing and philopatry (spawning site fidelity) of both species may also play a role in limiting the interaction of walleyes and saugers during the spawning season. Jennings et al. (1996) experimentally supported this hypothesis by demonstrating that natal homing in spawning walleyes is regulated by a genetically based response to environmental cues.

Despite reports of limited interactions between walleyes and saugers, our evaluation reconfirmed that walleyes in the Ohio River are experiencing introgression with walleye × sauger hybrids (White and Schell 1995). All F_1 hybrids that were identified had the Lake Erie (G) haplotype, which is consistent with saugeye broodstock sources (female Lake Erie walleye × male Ohio River sauger). Of the 33 F_X hybrids in our evaluation, 31 possessed the Lake Erie haplotype and two possessed the Ohio River (O) haplotype, suggesting that stocked fish are responsible for most of the documented introgression.

Because Lake Erie–strain walleyes are used for the propagation of saugeyes, the long-term impacts of introgression resulting from matings by these stocked saugeyes with either parental species in the Ohio River is a concern worth noting. Owing to its relatively low abundance, the recovering native walleye population may be particularly vulnerable to introgression from stocked saugeyes (Fiss et al. 1997). Annual stocking regimes are relatively constant, while native Ohio River walleye recruitment varies with ambient conditions. Recent tagging studies conducted on Ohio River saugers confirm that dams are not absolute barriers to fish movement (Chris Obara, personal communication).

Accuracy of Visual Identification

We have demonstrated that field identification of Ohio River percids is compromised by genetic introgression between walleyes and saugers. Misidentification in the field is a concern to managers collecting broodstock for stocking programs (Ward and Berry 1995; Billington et al. 1997). Because much broodstock collection has been done at night and genetic confirmation has not been done, many past field identifications may be questionable. Saugeyes (F_1 hybrids) were the most difficult to identify, only 75 out of a total sample of 210 being identified correctly. The remainder of this sample of 210 fish identified in the field as saugeyes were confirmed by allozyme analysis to be saugers (*n*

= 115), walleyes (n = 13), and F_X hybrids (n = 13). Walleyes were correctly identified more often in the field but still showed a 29% error (76 of 270). The other fish were determined by allozyme analysis to be saugers (n = 16), F_1 hybrids (n = 27), and F_X hybrids (n = 33). The implications could have negative consequences on a large scale if these fish were used as broodstock for propagating and stocking into areas where preservation of the genetic integrity of native stocks is a concern. Ryman and Utter (1987) have compiled several examples of how hybridization of cultured fish with wild fish can lead to the loss of coadapted gene complexes and genetic integrity, resulting in the loss of an entire species, subspecies, or unique population (Campton 1987).

In recent times, the Ohio River *Sander* fishery has been dominated by saugers, walleye abundance increasing in the most upstream reaches bordering Ohio. The Ohio River Recreational Use Survey (Schell et al. 1996) determined that the Ohio River *Sander* population was comprised of about 90% saugers and 5% each of walleyes and walleye × sauger hybrids (saugeyes). The fall 2002 percid electrofishing survey in Willow Island tailwater (Belleville Pool) yielded 527 (85.5%) saugers, 18 (3%) walleyes, and 71 (11.5%) walleye × sauger hybrids. These hybrids included 10 fish identified in the field as saugers that were later identified as saugeyes in the laboratory.

Several states have stocked saugeyes into the Ohio River drainage, including Pennsylvania and Ohio. West Virginia recently terminated its saugeye program owing to concerns about the genetic impacts of fertile hybrids (C. Obara, personal communication). The Ohio Department of Natural Resources' Division of Wildlife has a highly successful stocking program that includes stocking in 36 on-stream impoundments and 20 upground reservoirs as well as direct stockings into the Muskingum River (a tributary of the Ohio River). The proven advantages of walleye × sauger hybrids over either parental species in creating a quality fishery has been well documented in these altered and manmade habitats, many of which were previously stocked with Lake Erie–strain walleyes. Although the argument has been made that fisheries managers are justified in stocking saugeyes into riverine habitats that have been altered by dams and dredging, statutory and interjurisdictional responsibilities to protect native Ohio River walleye populations should also be considered. By agreement with the Great Lakes Fisheries Commission, stocking of this fertile hybrid in the Lake Erie drainage is limited to upground reservoirs.

Triploid saugeyes have been evaluated as an alternative to diploid saugeyes to reduce concerns about genetic introgression. However, a comparison of foraging performance (Czesny et al. 2002) confirmed that triploid juvenile saugeyes are less efficient than diploids, which could reduce the former's growth, increase their risk of predation, and decrease their survival after stocking. Propagation of a native strain of walleye for stocking in large rivers in the Ohio River drainage has not been evaluated. Walleye populations from the Pike Island and Hannibal pools contained high proportions of Ohio River haplotypes and would be the most suitable broodstock source. There is a strong possibility that this native strain is better adapted to the large tributaries of the Ohio River drainage and that as a result it would reproduce better than Lake Erie walleyes, perhaps providing a fishery equal to that provided by stocked saugeyes.

Management Recommendations

Management strategies to ensure the perpetuation of the unique Ohio River haplotype should be considered. Because there is currently no program for genetically confirming broodstock and introgression has compromised field identification, we recommend that the Belleville Pool (including the Willow Island and Devola tailwaters) and the Meldahl Pool not be used as percid broodstock sources. We also recommend establishment of a monitoring program at Ohio River tailwaters to document the occurrence and nature of hybrids and to further resolve the distribution and abundance of walleyes with the Ohio River haplotype. Development of a broodstock with that haplotype for stocking into the Ohio River and its major tributaries should be considered. Such a stocking program should be accompanied by genetic studies to evaluate the success of any Ohio River walleye stocking.

Acknowledgments

This work could not have been completed without the support of the Ohio River Fisheries Management Team consisting of Ohio, West Virginia, Indiana, Illinois, and Kentucky fisheries personnel. Laboratory assistance was provided by Natasha Silich and Jack Degner at the Illinois Natural History Survey. Funding for David Philipp and Todd Kassler's genetic analysis was provided by the Game Fish Committee of the Mississippi Interstate Cooperative Research Agreement. Laboratory as-

sistance at Ohio University was provided by Cy Stober, Elisa Weckerling, and Nancy Judd. Matt White's work was supported by the Ohio Department of Natural Resources, Division of Wildlife, through Federal Aid in Sport Fish Restoration Project F4DR10, the Department of Biological Sciences, and the Voinovich Center for Leadership and Public Affairs at Ohio University.

References

Allendorf, F. W., and L. W. Seeb. 2000. Concordance of genetic divergence among sockeye salmon populations at allozyme, nuclear DNA, and mitochondrial DNA markers. Evolution 54:640–651.

Alvarado Bremer, J. R., J. S. Bulak, and B. Ely. 1998. A polymerase chain reaction–restriction fragment length polymorphism (PCR–RFLP) assay for the discrimination of mitochondrial DNA from the Florida and northern subspecies of largemouth bass. Transactions of the American Fisheries Society 127: 507–511.

Arnaud, J. F., L. Madec, A. Guiller, and A. Bellido. 2001. Spatial analysis of allozyme and microsatellite DNA polymorphisms in the land snail *Helix aspersa* (Gastropoda: Helicidae). Molecular Ecology 10: 1563–1576.

Billington, N. 1996. Geographical distribution of mitochondrial DNA (mtDNA) variation in walleye, sauger, and yellow perch. Annals Zoology Fennici 33:699–706.

Billington, N., and P. D. N. Hebert. 1988. Mitochondrial DNA variation in Great Lakes walleye (*Stizostedion vitreum*) populations. Canadian Journal of Fisheries and Aquatic Sciences 45:643–654.

Billington, N., R. J. Barrette, and P. D. N. Hebert. 1992. Management implications of mitochondrial DNA variation in walleye stocks. North American Journal of Fisheries Management 12:276–284.

Billington, N., and R. M. Strange. 1995. Mitochondrial DNA analysis confirms the existence of a genetically divergent walleye population in northeastern Mississippi. Transactions of the American Fisheries Society 124:770–776.

Billington, N., R. C. Brooks, and R. C. Heidinger. 1997. Frequency of natural hybridization between saugers and walleyes in the Peoria Pool of the Illinois River, as determined by morphological and electrophoretic criteria. North American Journal of Fisheries Management 17:220–224.

Borer, S. O., L. M. Miller, and A. R. Kapuscinski. 1999. Microsatellites in walleye *Stizostedion vitreum*. Molecular Ecology 8:336–338.

Campton, D. E. 1987. Natural hybridization and introgression in fishes: methods of detection and genetic interpretations. Pages 161–192 *in* N. Ryman and F. Utter, editors. Population genetics and fishery management. University of Washington Press, Seattle.

Czesny, S., M. A. Garcia-Abiado, K. Dabrowski, P. Bajer, and M. Zalewski. 2002. Comparison of foraging performance of diploid and triploid saugeyes (sauger × walleye). Transactions of the American Fisheries Society 131:980–985.

Estoup, A., F. Rousset, Y. Michalakis, J. M. Cornuet, M. Adriamange, and R. Guyomard. 1998. Comparative analysis of microsatellite and allozyme markers: a case study investigating microgeographic differentiation in brown trout (*Salmo trutta*). Molecular Ecology 7:339–353.

Fields, R. D., K. R. Johnson, and G. H. Thorgaard. 1989. DNA fingerprinting in rainbow trout detected by hybridization with DNA bacteriophage M13. Transactions of the American Fisheries Society 118:78–81.

Fiss, F. C., S. M. Sammons, P. W. Bettoli, and N. Billington. 1997. Reproduction among saugeyes (F_X hybrids) and walleyes in Normandy Reservoir, Tennessee. North American Journal of Fisheries Management 17:215–219.

Flint, R. F. 1971. Glacial and quaternary geology. Wiley, New York.

Gatt, M. H., M. M. Ferguson, and A. P. Liskauskas. 2000. Comparison of control region sequencing and fragment RFLP analysis for resolving mitochondrial DNA variation and phylogenetic relationships among Great Lakes walleye. Transactions of the American Fisheries Society 129:1288–1299.

Gatt, M. H., D. J. Fraser, A. P. Liskauskas, and M. M. Ferguson. 2002. Mitochondrial DNA variation and stock structure of walleye from eastern Lake Huron: an analysis of contemporary and historical samples. Transactions of the American Fisheries Society 131: 99–108.

Goudet, J. 1995. FSTAT, version 1.2: a computer program to calculate *F*-statistics. Journal of Heredity 86:485–486.

Hillis, D. M., B. K. Mable, A. Larson, S. K. Davis, and E. A. Zimmer. 1996. Nucleic acids IV: sequencing and cloning. Pages 321–384 *in* D. M. Hillis, C. Moritz, and B. K. Mable, editors. Molecular systematics. Sinauer Associates, Sunderland, Massachusetts.

IUBMB (International Union of Biochemistry and Molecular Biology). 1992. Enzyme nomenclature, 1992. Academic Press, San Diego, California.

Jennings, M. J., J. E. Claussen, and D. P. Philipp. 1996. Evidence for heritable preferences for spawning habitat between two walleye groups. Transactions of the American Fisheries Society 125:978–982.

Kassler, T. W., and D. P. Philipp. 2000. Genetic assessment of two *Stizostedion* species and their hybrids. Illinois Natural History Survey, Center for Aquatic Ecology, Final Report for the Ohio Department of Natural Resources, Division of Wildlife, Athens.

Kassler, T. W., and D. P. Philipp. 2001. Genetic assessment of two *Stizostedion* species and their hybrids. Illinois Natural History Survey, Center for Aquatic Ecology, Final Report for the Ohio Department of Natural Resources, Division of Wildlife, Athens.

King, T. L., S. T. Kalinowski, W. B. Schill, A. P. Spidle, and B. A. Lubinski. 2001. Population structure of Atlantic salmon (*Salmo salar* L.): a range-wide per-

spective from microsatellite DNA variation. Molecular Ecology 10:807–821.

Kocher, T. D., W. K. Thomas, A. Meyer, S. V. Edwards, S. Paabo, X. F. Villablanca, and A. C. Wilson. 1989. Dynamics of mitochondrial DNA evolution in animals: amplification and sequencing with conserved primers. Proceedings of the National Academy of Sciences of the USA 86:6196–6200.

Koppelman, J. B., and D. P. Philipp. 1986. Genetic applications in muskellunge management. Pages 111–121 *in* G. E. Hall, editor. Managing muskies. American Fisheries Society, Special Publication 15, Bethesda, Maryland.

Meyer, A., T. D. Kocher, and P. Basasibwaki. 1990. Monophytic origin of Lake Victoria cichlid fishes suggested by mitochondrial DNA sequences. Nature (London) 347:550–563.

Nielsen, J. L., K. D. Crow, and M. C. Fountain. 1999. Microsatellite diversity and conservation of a relic trout population: McCloud River redband trout. Molecular Ecology 8:S129–S142.

Palmer, G. C. 1999. Genetic characterization and intermixing of walleye stocks in Claytor Lake and the upper New River, Virginia. Master's thesis. Virginia Polytechnic Institute and State University, Blacksburg.

Park, S. D. E. 2001. Trypanotolerance in West African cattle and the population genetic effects of selection. Doctoral dissertation. University of Dublin, Dublin, Ireland.

Philipp, D. P., W. F. Childers, and G. S. Whitt. 1979. Evolution of differential patterns of gene expression: a comparison of the temporal and spatial patterns of isozyme locus expression in two closely related fish species (northern largemouth bass, *Micropterus salmoides salmoides*, and smallmouth bass, *Micropterus dolomieui*). Journal of Experimental Zoology 210:473–488.

Philipp, D. P., J. E. Claussen, T. W. Kassler, and J. M. Epifanio. 2002. Conservation genetics and black bass management. Pages 349–363 *in* D. P. Philipp and M. S. Ridgway, editors. American Fisheries Society, Symposium 31, Bethesda, Maryland.

Raymond, M., and F. Rousset. 1995. GENEPOP, version 1.2: population genetics software for exact tests and ecumenicism. Journal of Heredity 86:248–249.

Rice, W. R. 1989. Analyzing tables of statistical tests. Evolution 43:223–225.

Rogers, J. S. 1972. Measures of genetic similarity and genetic distance. Studies in Genetics, University of Texas Publications 7213:145–153.

Ryman, N., and F. Utter, editors. 1987. Population genetics in fishery management. University of Washington Press, Seattle.

Schell, S. A., D. J. Bright, J. A. Marshall, and M. A. Greenlee. 1996. Ohio River recreational use survey, 1992–1993. Ohio Department of Natural Resources, Division of Wildlife, Federal Aid in Sport Fish Restoration, Project F-69-P, Final Report, Athens.

Schneider, S., D. Roessli, and L. Excoffier. 2000. Arlequin, version 2.0: a software for population genetics data analysis. Genetics and Biometry Laboratory, University of Geneva, Geneva, Switzerland.

Shaklee, J. B., F. W. Allendorf, D. C. Morizot, and G. S. Whitt. 1990. Gene nomenclature for protein-coding loci in fish. Transactions of the American Fisheries Society 119:2–15.

Sneath, P. H. A., and R. R. Sokal. 1973. Numerical taxonomy. Freeman, San Francisco.

Stepien, C. A., and J. E. Faber. 1998. Population genetic structure, phylogeography, and spawning philopatry in walleye (*Stizostedion vitreum*) from mitochondrial DNA control region sequences. Molecular Ecology 7:1757–1776.

Swofford, D. L., and R. B. Selander. 1981. BIOSYS-1: a FORTRAN program for the comprehensive analysis of electrophoretic data in population genetics and systematics. Journal of Heredity 72:281–283.

Thornbury, W. D. 1965. Regional geomorphology of the United States. Wiley, New York.

Todd, T. N., and R. C. Hass. 1993. Genetic and tagging evidence for movement of walleye between Lake Erie and Lake St. Clair. Journal of Great Lakes Research 19:445–452.

Trautman, M. B. 1981. The fishes of Ohio. Ohio State University Press, Columbus.

Ward, N. E., and C. R. Berry, Jr. 1995. Evaluation of skin pigmentation for identifying adult saugers and walleye–sauger F_1 hybrids collected from Lake Sakakawea, North Dakota. Progressive Fish-Culturist 57:302–304.

White, M. M., and S. Schell. 1995. An evaluation of the genetic integrity of Ohio River walleye and sauger stocks. Pages 52–60 *in* J. L. Schramm, Jr., and R. G. Piper, editors. Uses and effects of cultured fishes in aquatic ecosystems. American Fisheries Society, Symposium 15, Bethesda, Maryland.

Wirth, T., R. Saint-Laurent, and L. Bernatchez. 1999. Isolation and characterization of microsatellite loci in the walleye (*Stizostedion vitreum*) and cross-species amplification within the family Percidae. Molecular Ecology 8:1960–1962.

Appendix: Microsatellite Allele Frequencies and Genetic Variability

TABLE A.1.—Estimates of microsatellite allele frequencies and measures of genetic variability for populations of walleye from the Ohio River, Pymatuning Lake, and Salt Fork Lake. Abbreviations are as follows: McA, McAlpine; Mar, Markland; Meh, Mehldahl; Gre, Greenup; Rac, Racine; Wil, Willow Island; Pike, Pike Island; NC, New Cumberland; Pym, Pymatuning Lake; Salt Fork Lake; H_o, observed heterozygosity; H_e, Hardy–Weinberg expected heterozygosity, and A, mean number of alleles. The letters G and O indicate the Lake Erie and Ohio River haplotypes, respectively.

Locus and summary statistics	Allele	Population					
		McAG	MarG	MehG	GreG	ByrdG	RacG
Svi4							
	105	0.000	0.000	0.000	0.000	0.000	0.000
	107	0.071	0.031	0.000	0.416	0.200	0.000
	109	0.285	0.125	0.000	0.083	0.200	0.000
	111	0.428	0.312	0.210	0.166	0.500	0.000
	113	0.071	0.218	0.290	0.250	0.100	0.000
	115	0.000	0.312	0.390	0.097	0.000	0.250
	117	0.142	0.000	0.100	0.277	0.000	0.187
	119	0.000	0.000	0.000	0.041	0.000	0.562
	121	0.000	0.000	0.000	0.028	0.000	0.000
	125	0.000	0.000	0.000	0.014	0.000	0.000
Svi6							
	123	0.000	0.000	0.000	0.000	0.100	0.000
	127	0.000	0.000	0.000	0.000	0.100	0.000
	129	0.000	0.000	0.000	0.000	0.100	0.000
	131	0.000	0.000	0.184	0.062	0.200	0.187
	133	0.000	0.071	0.263	0.156	0.200	0.062
	135	0.285	0.071	0.052	0.093	0.300	0.312
	137	0.142	0.178	0.237	0.218	0.000	0.187
	139	0.142	0.142	0.000	0.234	0.000	0.250
	141	0.375	0.464	0.184	0.297	0.000	0.000
	143	0.071	0.000	0.026	0.031	0.000	0.000
	145	0.000	0.071	0.052	0.046	0.000	0.000
	153	0.000	00.00	0.000	0.000	0.000	0.000
Svi17							
	101	0.000	0.000	0.026	0.000	0.000	0.000
	105	0.000	0.000	0.000	0.000	0.000	0.000
	107	0.000	0.000	0.052	0.013	0.000	0.000
	109	0.000	0.205	0.000	0.079	0.100	0.000
	111	0.285	0.147	0.394	0.131	0.000	0.125
	113	0.142	0.291	0.315	0.394	0.400	0.437
	115	0.000	0.176	0.105	0.013	0.100	0.437
	117	0.571	0.176	0.026	0.223	0.400	0.000
	119	0.000	0.000	0.052	0.131	0.000	0.000
	121	0.000	0.000	0.026	0.013	0.000	0.000
	123	0.000	0.000	0.000	0.000	0.000	0.000
Svi18							
	118	0.000	0.000	0.000	0.136	0.000	0.000
	120	0.000	0.000	0.184	0.151	0.000	0.000
	122	0.071	0.156	0.315	0.348	0.000	0.125
	124	0.571	0.062	0.500	0.198	0.300	0.500
	126	0.214	0.437	0.000	0.151	0.200	0.375
	128	0.142	0.218	0.000	0.000	0.500	0.000
	130	0.000	0.125	0.000	0.015	0.000	0.000
	132	0.000	0.000	0.000	0.000	0.000	0.000
	134	0.000	0.000	0.000	0.000	0.000	0.000
SviL5							
	184	0.000	0.050	0.000	0.000	0.000	0.000
	186	0.000	0.050	0.000	0.000	0.000	0.000
	188	0.000	0.000	0.000	0.000	0.000	0.000
	190	0.000	0.000	0.000	0.016	0.000	0.000
	192	0.071	0.000	0.000	0.100	0.000	0.000
	194	0.071	0.000	0.139	0.200	0.200	0.000
	196	0.071	0.100	0.111	0.250	0.100	0.500
	198	0.286	0.250	0.194	0.150	0.300	0.375
	200	0.071	0.350	0.055	0.000	0.200	0.000
	202	0.000	0.000	0.222	0.100	0.000	0.125
	204	0.071	0.250	0.083	0.067	0.200	0.000
	206	0.357	0.000	0.111	0.083	0.000	0.000

TABLE A1.—Extended.

Locus and summary statistics	Population							
	WilG	WilO	PikeIG	PikeIO	NCG	NCO	PymG	SaltG
Svi4								
	0.000	0.000	0.000	0.000	0.020	0.000	0.167	0.000
	0.000	0.000	0.000	0.083	0.020	0.000	0.083	0.000
	0.235	0.000	0.125	0.250	0.110	0.178	0.000	0.125
	0.323	0.625	0.250	0.138	0.250	0.462	0.333	0.500
	0.412	0.375	0.312	0.305	0.270	0.142	0.208	0.250
	0.000	0.000	0.062	0.055	0.180	0.107	0.083	0.000
	0.029	0.000	0.125	0.000	0.130	0.107	0.125	0.125
	0.000	0.000	0.125	0.167	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Svi6								
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	0.033	0.000	0.000	0.000	0.045	0.037	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	0.100	0.375	0.000	0.000	0.454	0.537	0.083	0.125
	0.033	0.000	0.083	0.093	0.136	0.107	0.416	0.000
	0.300	0.375	0.166	0.187	0.159	0.107	0.333	0.375
	0.133	0.250	0.083	0.093	0.113	0.107	0.083	0.125
	0.033	0.000	0.083	0.093	0.091	0.037	0.000	0.125
	0.266	0.000	0.083	0.093	0.000	0.074	0.083	0.250
	0.033	0.000	0.500	0.437	0.000	0.000	0.000	0.000
	0.033	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	0.033	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Svi17								
	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	0.071	0.125	0.000	0.000	0.000	0.000	0.250	0.000
	0.000	0.000	0.000	0.000	0.022	0.000	0.083	0.000
	0.035	0.125	0.000	0.000	0.022	0.357	0.167	0.000
	0.428	0.250	0.125	0.277	0.477	0.392	0.250	0.625
	0.107	0.000	0.500	0.389	0.227	0.035	0.167	0.125
	0.107	0.375	0.250	0.111	0.022	0.143	0.083	0.000
	0.142	0.000	0.062	0.222	0.136	0.070	0.083	0.250
	0.071	0.000	0.062	0.000	0.000	0.000	0.041	0.000
	0.000	0.013	0.000	0.000	0.000	0.000	0.000	0.000
	0.035	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Svi18								
	0.000	0.000	0.000	0.055	0.000	0.000	0.083	0.000
	0.000	0.000	0.000	0.110	0.046	0.143	0.083	0.125
	0.125	0.000	0.062	0.305	0.022	0.107	0.291	0.125
	0.187	0.250	0.375	0.083	0.432	0.393	0.166	0.500
	0.343	0.500	0.250	0.166	0.295	0.214	0.166	0.125
	0.281	0.125	0.250	0.194	0.159	0.071	0.041	0.125
	0.062	0.125	0.000	0.083	0.046	0.071	0.125	0.000
	0.000	0.000	0.062	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000	0.041	0.000
SviL5								
	0.000	0.000	0.000	0.000	0.000	0.000	0.083	0.000
	0.000	0.125	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.125	0.000	0.000	0.000	0.083	0.000
	0.000	0.000	0.000	0.028	0.000	0.000	0.000	0.000
	0.117	0.000	0.062	0.028	0.000	0.071	0.167	0.000
	0.088	0.125	0.000	0.111	0.023	0.000	0.041	0.125
	0.294	0.250	0.000	0.222	0.363	0.178	0.000	0.250
	0.147	0.000	0.000	0.194	0.046	0.036	0.292	0.000
	0.117	0.000	0.125	0.000	0.046	0.357	0.000	0.000
	0.088	0.000	0.312	0.083	0.250	0.143	0.250	0.250
	0.059	0.125	0.125	0.055	0.046	0.143	0.000	0.125
	0.088	0.125	0.125	0.083	0.046	0.071	0.083	0.125

TABLE A.1.—Continued.

Locus and summary statistics	Allele	Population					
		McAG	MarG	MehG	GreG	ByrdG	RacG
	208	0.000	0.000	0.083	0.000	0.000	0.000
	210	0.000	0.000	0.000	0.000	0.000	0.000
	214	0.000	0.000	0.000	0.033	0.000	0.000
H_o		0.729	0.758	0.760	0.804	0.782	0.670
H_e		0.600	0.576	0.456	0.543	0.680	0.450
A		4.8	5.4	6.0	8.0	4.4	3.4

TABLE A1.—Extended Continued.

Locus and summary statistics	Population							
	WilG	WilO	PikeIG	PikeIO	NCG	NCO	PymG	SaltG
	0.000	0.000	0.000	0.194	0.159	0.000	0.000	0.125
	0.000	0.250	0.125	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.023	0.000	0.000	0.000
H_o	0.787	0.764	0.793	0.806	0.753	0.750	0.785	0.821
H_e	0.588	0.450	0.275	0.311	0.372	0.500	0.650	0.533
A	7.0	4.0	5.6	6.6	6.8	6.0	4.6	6.8